

## Four problems — answers

Item 1 is worked in full on the handout.

### 2 — The same mean, made in bursts

$$\text{Fano} \approx 1 + b = 5, \quad \eta^2 = \frac{5}{25} = 0.20, \quad \eta = 0.447.$$

More than twice item 1's 0.20, at the same mean. **What changed that the mean cannot see** is the size of each arrival. Item 1's proteins arrive one at a time; these arrive four at a time, from a quarter as many transcripts. The mean is the product of how often and how many, so it cannot tell the two apart. The spread can.

The design rule this gives you: at a fixed mean, a strong promoter with a weak RBS is quieter than a weak promoter with a strong RBS. A promoter library and an RBS library that reach the same expression level are not interchangeable.

**Which to build for 25 per cell with the least variation:** raise the promoter and weaken the RBS until  $b$  is near 1. At  $b = 1$  the Fano factor is about 2 and  $\eta = \sqrt{2/25} = 0.28$ ; the floor, as  $b$  falls toward zero, is the Poisson's 0.20. Going lower than that takes feedback, which is item 3.

The exact two-stage result is  $\text{Fano} = 1 + k_p/(\gamma_m + \gamma_p)$ , which is  $1 + b$  only when the mRNA lifetime is much shorter than the protein's. For the  $b = 10$  pair on the slides, with mRNA ten times shorter-lived, that formula gives **10.09** against  $1 + b = 11$ , and the simulation gave **10.3**. The same correction applies here: at  $b = 4$  with the same lifetime ratio the exact value is 4.64, not 5, so  $\eta$  is nearer 0.43 than 0.45. Either is a full answer — the handout says the mRNA is short-lived, which is the regime where  $1 + b$  is meant to hold.

### 3 — The same gene, repressing itself

$$g = \frac{3 \times 1}{1 + 1} = 1.5, \quad \text{Fano} = \frac{1}{1 + 1.5} = 0.4$$

$$\eta_{\text{constitutive}} = \frac{1}{\sqrt{100}} = 0.100, \quad \eta_{\text{self-repressed}} = \sqrt{\frac{0.4}{100}} = 0.063$$

Cutting the Fano factor to 0.4 cuts  $\eta$  by  $\sqrt{2.5} = 1.58$ . The slide showed  $n = 1, 2, 4$  and 8;  $n = 3$  falls between them, as it should.

**What the feedback changed.** Not the event rate: at the same mean, births and deaths each happen at  $\gamma x$  per unit time with or without feedback, so the noise going *in* is identical. What changed is how fast a fluctuation is pulled back. A cell that drifts high makes less, so it returns at rate  $\gamma(1 + g)$  instead of  $\gamma$ . Each

fluctuation lasts a shorter time, and the variance is the noise going in divided by twice that return rate. It is the same mechanism that made the self-repressed gene respond faster in session 7.

Two limits. The result is for proteins made one at a time; with bursts the benefit is smaller. And the result is about intrinsic noise only. Slow extrinsic changes in the gene's own rates are buffered too, and more strongly: at steady state  $\gamma x = f(x)$ , so  $d \ln x^* / d \ln \beta = 1/(1 + g)$ , and a slow wobble in the promoter strength reaches  $x$  divided by  $1 + g$  in CV, not by  $\sqrt{1 + g}$ .

## 4 — Reading Table 1

There is more than one good answer. Three comparisons that each tell you something:

- **RP22 with and without IPTG.** Same strain, inducer only. Intrinsic noise falls from 25 to 6.3 as expression rises about thirtyfold (intensity  $0.030 \rightarrow 1.00$ ). That is the  $1/\langle n \rangle$  law: more transcripts, smaller relative spread. Extrinsic noise falls too, from 33 to 9.8, because saturating IPTG makes the cell-to-cell variation in LacI irrelevant (the paper's own explanation, note 13, p. 1186).
- **M22 against M22 + repressilator.** Extrinsic noise jumps from 5.4 to 42. The clock moves LacI up and down in every cell, out of phase between cells, so both promoters swing together, which is what extrinsic means. Be careful how much of that jump you credit to the clock: partial repression alone puts  $\eta_{\text{ext}}$  in the thirties (RP22 at intensity 0.030 reads 33, MG22 at 0.057 reads 32), and note 13 on p. 1186 says  $\eta_{\text{ext}}$  peaks at intermediate induction. The repressilator strain sits at intensity 0.18, which is in that range. Intrinsic noise is 12, but the mean is lower too (intensity 0.18), so compare it against the paper's own fit to strain M22,  $\eta_{\text{int}}^2 \approx c_1/m + c_2$  with  $c_1 = 7 \times 10^{-4}$  and  $c_2 = 3 \times 10^{-3}$  (Fig. 3B caption, p. 1185). At  $m = 0.18$  that predicts  $\eta_{\text{int}} = 8.3 \times 10^{-2}$ , so the measured 12 is about 1.5 times higher: noise is larger while a gene is being switched than at a steady state (p. 1186). Use the Fig. 3B constants, not Fig. 3C's — that fit is for the noisier strain D22.
- **M22 against RP22.** Repressing the promoter about thirtyfold raises intrinsic noise from 5.5 to 25 and extrinsic from 5.4 to 33. The intrinsic rise is the count law again; the extrinsic rise is LacI, which varies from cell to cell and which the constitutive strain does not have at all.

**The check**, on all four rows. M22:  $5.5^2 + 5.4^2 = 59.4$ ,  $\sqrt{\phantom{x}} = 7.71$  against 7.7. RP22:  $25^2 + 33^2 = 1714$ ,  $\sqrt{\phantom{x}} = 41.4$  against 41. RP22 + IPTG:  $6.3^2 + 9.8^2 = 135.7$ ,  $\sqrt{\phantom{x}} = 11.65$  against 11.7. Repressilator:  $12^2 + 42^2 = 1908$ ,  $\sqrt{\phantom{x}} = 43.7$  against 43 — the worst of the four at 1.6%, and still inside the 39–47 confidence interval the table prints for it. The table rounds; the identity holds.

## Appendix — the mean and the variance, without solving for $P_n$

This is the question the room asks every year, and it has a clean answer that needs none of the ladder. Start from the master equation,

$$\frac{dP_n}{dt} = kP_{n-1} + \gamma(n+1)P_{n+1} - (k + \gamma n)P_n,$$

multiply by  $n$ , and sum over all  $n$ . Shift the index in each sum so that  $P_m$  appears throughout:

$$\sum_n n k P_{n-1} = k(\langle n \rangle + 1), \quad \sum_n n \gamma(n+1) P_{n+1} = \gamma(\langle n^2 \rangle - \langle n \rangle).$$

Subtract the two loss terms,  $k\langle n \rangle$  and  $\gamma\langle n^2 \rangle$ , and everything with an  $\langle n^2 \rangle$  in it cancels:

$$\frac{d\langle n \rangle}{dt} = k - \gamma\langle n \rangle \quad \implies \quad \langle n \rangle = \frac{k}{\gamma}.$$

That is session 5's deterministic equation, recovered as a statement about an average rather than assumed.

Do the same with  $n^2$ . The  $\langle n^3 \rangle$  terms cancel for the same reason, leaving

$$\frac{d\langle n^2 \rangle}{dt} = 2k\langle n \rangle + k - 2\gamma\langle n^2 \rangle + \gamma\langle n \rangle,$$

and at steady state, with  $\langle n \rangle = k/\gamma$ ,

$$\langle n^2 \rangle = \left(\frac{k}{\gamma}\right)^2 + \frac{k}{\gamma}, \quad \sigma^2 = \langle n^2 \rangle - \langle n \rangle^2 = \frac{k}{\gamma} = \langle n \rangle.$$

**Why this matters more than the arithmetic.** The equation for  $\langle n \rangle$  involved only  $\langle n \rangle$ , and the equation for  $\langle n^2 \rangle$  involved only  $\langle n \rangle$  and  $\langle n^2 \rangle$ . The hierarchy *closed*. That happens because birth and death are **linear** in  $n$ . Put a Hill function in — which is what item 3 does — and the equation for  $\langle x \rangle$  contains  $\langle f(x) \rangle$ , which is not  $f(\langle x \rangle)$ , and the equation for that contains something worse. There is no exact answer to read off, for item 3 or for almost any real circuit. That is the honest reason the feedback result is obtained from the linear-noise approximation and then checked against simulation, rather than derived.